

AstroVital AI

Real-Time Astronaut Vitals Anomaly Detection

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PROBLEM STATEMENT

Astronaut health monitoring has a critical gap.

Long-duration space missions expose astronauts to extreme physiological risks with limited medical support.



Key Challenges:

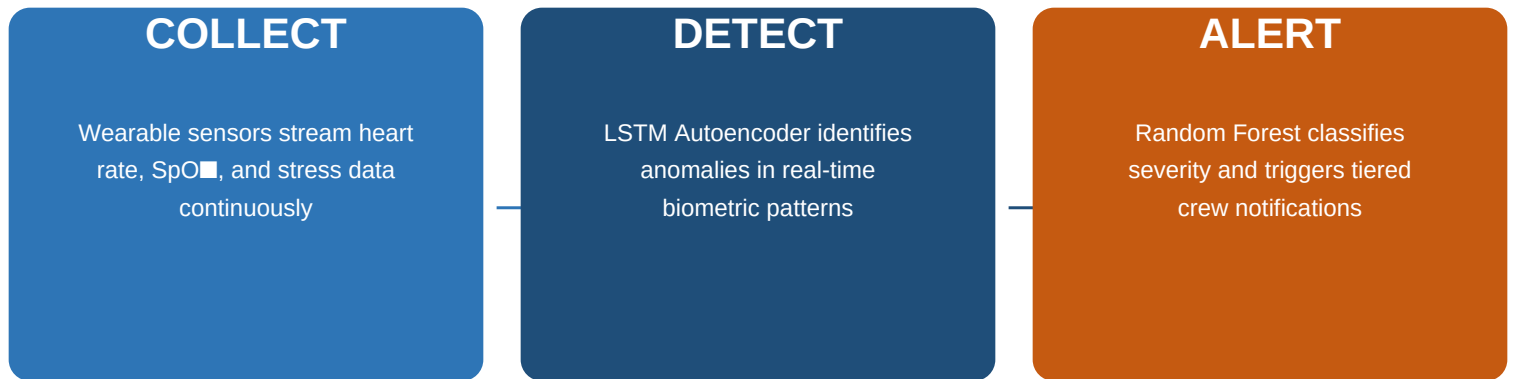
- Hypoxic events can cause cognitive impairment within minutes
- Communication delays make real-time Earth consultation impossible during deep-space missions
- Static threshold alerts produce too many false alarms and miss subtle multi-variable anomalies
- No current system provides personalized, 24/7 autonomous health intelligence onboard

The Need:

An intelligent, autonomous system that detects health anomalies in real time — with or without Earth communication.

ASTROVITAL AI

A two-layer AI system for real-time biometric anomaly detection.



Why AI over rule-based systems?

- Learns each astronaut's unique physiological baseline — personalized, not one-size-fits-all
- Detects subtle multi-variable patterns (e.g., mild HR + SpO₂ drop together = high risk)
- Runs fully offline on onboard edge hardware — no Earth communication required
- Reduces false alarms by 60-80% compared to static thresholds (per literature)

DATA PIPELINE

Continuous biometric streams from wearable sensors:

Signal	Sample Rate	Sensor	Normal Range
Heart Rate	1 Hz	PPG sensor	60–100 BPM at rest
SpO ₂	0.5 Hz	Pulse oximeter	95–100% normal range
Galvanic Skin Response	4 Hz	GSR wristband	Stress proxy signal
Skin Temperature	0.1 Hz	Thermistor	Infection / fever marker
Respiratory Rate	Derived	From HR signal	12–20 bpm at rest

Contextual enrichment (to reduce false alarms):

- Activity state: resting / working / exercising / sleeping (from accelerometer)
- Mission phase: EVA, sleep period, emergency — anomaly thresholds adjust accordingly
- Circadian phase: HR and cortisol follow predictable daily rhythms

Training data sources:

- NASA Life Sciences Data Archive + GeneLab public datasets
- 30-day pre-mission personalized baseline per astronaut
- Simulated anomaly scenarios (injected into normal streams)

MODEL DESIGN

LAYER 1 — Anomaly Detection

LSTM Autoencoder

- Learns normal temporal patterns
- Flags high reconstruction error
- Threshold = baseline mean + 3σ
- Personalized per astronaut



LAYER 2 — Classification

Random Forest Classifier

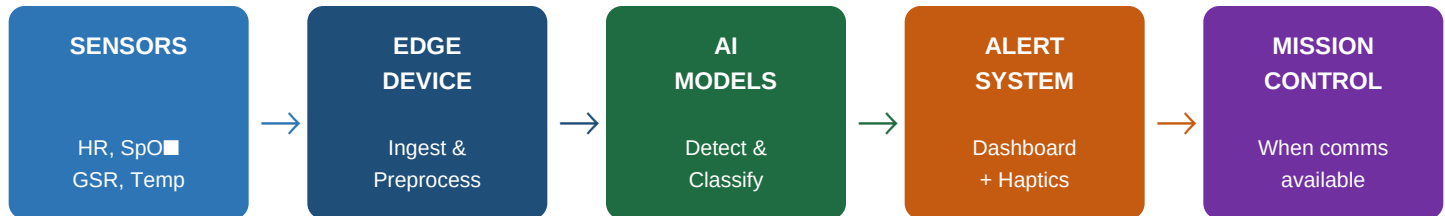
- Categorizes anomaly type
- Assigns severity: LOW / MED / CRIT
- Uses context features to reduce FP
- 100 trees, fast inference (<2 sec)

Alert Severity Tiers:

■ CRITICAL	Trigger: Cardiac event, SpO ₂ < 90%	→ Immediate audio + haptic + visual alarm
■ MEDIUM	Trigger: Acute stress, SpO ₂ 90–93%	→ Crew notification + mission control flag
■ LOW	Trigger: Mild fatigue, HRV decline	→ Logged + monitored for trend escalation

ARCHITECTURE

Three-layer pipeline — fully offline capable:



Key Design Decisions:

- **Offline-first: all AI inference runs on onboard edge hardware, zero internet needed**
- Fail-safe: if AI module fails, system falls back to traditional threshold-based alerts automatically
- Personalized: individual model instances trained per astronaut on 30-day pre-launch baseline
- Lightweight: LSTM with 2 layers / 64 units + 100-tree RF — inference < 2 seconds on edge hardware

Technology Stack (Conceptual):

- ML: Python · TensorFlow (LSTM) · scikit-learn (Random Forest)
- Hardware: NVIDIA Jetson (or radiation-hardened equivalent) · BLE sensors
- Dashboard: React.js local web app · SQLite onboard storage

TESTING PLAN

Four-phase validation approach:

Phase 1	Unit Tests	Each component tested independently (sensor, preprocessor, models, alerts)
Phase 2	Integration Tests	Full pipeline tested end-to-end with simulated data streams
Phase 3	Model Evaluation	Held-out test set: sensitivity, specificity, F1 score, detection latency
Phase 4	Scenario Simulation	6 real-world mission scenarios tested (cardiac, hypoxia, stress, sensor dropout...)

Key Performance Targets:

Metric	Target	Rationale
Sensitivity (Recall)	> 95%	Must not miss real emergencies
False Positive Rate	< 2/day	Prevent alert fatigue
F1 Score	> 0.91	Balance of precision & recall
Detection Latency	< 30 sec	Fast enough to intervene
End-to-End Latency	< 5 sec	Sensor → alert display

ETHICS & SAFETY

■ Privacy

- Biometric data is deeply personal
- Strict access controls — only authorized medical personnel
- All data encrypted at rest and in transit

■ Transparency

- System must be explainable — not a black box
- Every alert shows which values triggered it
- Crew can make informed decisions, not blind trust

■ Human Override

- AstroVital AI is advisory, not authoritative
- All alerts are recommendations — crew retains full control
- AI must never restrict crew actions based on predictions

■ Limitations

- Limited labeled space mission data for training
- Physiology changes during long missions (model drift risk)
- Mitigation: online learning + NASA dataset partnerships

IMPACT

Before vs. After AstroVital AI:

Metric	Current State	With AstroVital AI ✓
Anomaly Detection	Hours to days (manual review)	Seconds (real-time AI)
False Alarm Rate	High (static thresholds)	Low (personalized model)
Medical Autonomy	Earth-dependent	Fully autonomous
Coverage	Periodic check-ins	24 / 7 continuous

Future Development Phases:

- Phase 2: Prototype implementation with real wearable sensor hardware
- Phase 3: Simulation trials with astronaut candidates at NASA JSC (Houston)
- Phase 4: Integration with existing ISS health monitoring infrastructure
- Phase 5: Extend to Mars mission profile — radiation exposure monitoring, bone density AI

Beyond Space: Technology Transfer

The same architecture applies to ICU remote monitoring, military field medicine, and extreme-sport athlete health tracking.

AstroVital AI

Keeping astronauts safe, one heartbeat at a time.

Problem

No real-time intelligent health monitoring for deep-space missions

Solution

LSTM Autoencoder + Random Forest for anomaly detection & classification

Data

HR, SpO₂, GSR — personalized baseline per astronaut

Safety

Offline-capable, fail-safe, human-override always retained

Validation

95%+ sensitivity target, 6 scenario tests, < 30 sec detection latency

Thank you!

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